

Mengchen Fan

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SUMMARY

- My research develops AI methodologies for learning from complex and heterogeneous data. I focus on efficient and reliable models, including model compression and adaptation, multimodal data fusion, longitudinal-data reasoning, and distributed learning. Medical imaging is a primary application area for my research.
- **Research Interests:** Medical AI, Distributed Learning, and Federated Learning.
- **Teaching Interests:** Machine Learning, Deep Learning, Programming, and Algorithms.

EDUCATION

- **University of Alabama at Birmingham** Aug 2022 – Present
Ph.D. in Computer Science, GPA: 4.0/4.0; Expected May 2027 Birmingham, AL, United States
Advisor: Baocheng Geng, Ph.D.
Dissertation Research (provisional): *Medical AI for Longitudinal Imaging and Reasoning*
- **Syracuse University** Sept 2017 – May 2019
M.S. in Computer Engineering, GPA: 3.867/4.0 Syracuse, NY, United States
- **North China Electric Power University** Sept 2012 – May 2016
B.S. in Electrical Engineering, GPA: 3.47/4.0 China

RESEARCH EXPERIENCE

- **University of Alabama at Birmingham** Aug 2022 – Present
Doctoral Researcher in Computer Science Birmingham, AL, United States
 - Developed region-aware distillation methods that produced lightweight 3D medical-image segmentation models.
 - Developing longitudinal MRI/CT reasoning benchmarks and agent-based approaches toward report generation.
 - Developed federated and decentralized learning methods for heterogeneous clients through personalized training and representative knowledge sharing.
 - **Outcomes:** Ten peer-reviewed publications, including seven first or co-first author papers. The MICCAI 2026 paper received an early-accept decision (top 9% of 4,601 submissions) and was selected for a Spotlight presentation. My work on longitudinal brain MRI reasoning was also accepted to NeurIPS 2026.
- **Pulin Che Laboratory, UAB Heersink School of Medicine** May – Aug 2026
Research Intern Birmingham, AL, United States
 - Developed deep learning pipelines for ImageStreamX imaging flow cytometry to quantify pathway-specific Golgi morphology.
 - Developed interpretable classifiers and validation methods for inflammatory activation, signaling-pathway identification, and generalization across species and unseen compounds.
 - **Outcomes:** Three submitted manuscripts on Golgi morphology analysis, signaling-pathway identification, and drug repurposing.

PUBLICATIONS

J–Journal, C–Conference, A–Abstract, P–Patent

[†]Equal contribution

- [C.1] Qizhen Lan[†], **Mengchen Fan**[†], Hang Zhang, Jingwei Duan, Moule Lin, Jialin Chen, Baocheng Geng, and Xiaoqian Jiang. *BrainTRACE: Tracing Longitudinal, Multimodal, and Volumetric Evidence in Brain MRI Clinical Reasoning*. **Conference on Neural Information Processing Systems (NeurIPS)**, 2026.
- [C.2] **Mengchen Fan**, Baocheng Geng, Xi Xiao, Tianyang Wang, Siyuan Mei, Pulin Che, Xiaoqian Jiang, and Qizhen Lan. *Detail Consistent Stage-Wise Distillation for Efficient 3D MRI Segmentation*. **International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)**, 2026, **Early Accept (top 9%); Spotlight**.
- [C.3] **Mengchen Fan**, Fei Zhao, Chengcui Zhang, Shuya Feng, and Baocheng Geng. *On Optimal Modality Compression for Bandwidth Constrained Multimodal Fusion*. **IEEE International Conference on Information Reuse and Integration for Data Science (IRI)**, 2026.
- [C.4] **Mengchen Fan**, Tianyun Zhang, and Baocheng Geng. *A Unified Framework for the Convergence and Weight Pruning in Federated Learning*. **ACM Multimedia Asia**, 2025.
- [C.5] **Mengchen Fan**, Keren Li, Tianyun Zhang, Qing Tian, and Baocheng Geng. *PFedDST: Personalized Federated Learning with Decentralized Selection Training*. **IEEE International Joint Conference on Neural Networks (IJCNN)**, 2025.
- [J.1] **Mengchen Fan**, Tianyun Zhang, Xiaolong Ma, Jiacheng Guo, Zheng Zhan, Shanglin Zhou, Minghai Qin, Fei Sun, Yen-Kuang Chen, Caiwen Ding, Makan Fardad, and Yanzhi Wang. *A Unified DNN Weight Compression Framework Using Reweighted Optimization Methods*. **Intelligent Systems with Applications**, 2025, vol. 27.
- [J.2] Joseph Casey, Qianjiao Chen, **Mengchen Fan**, Baocheng Geng, Roman Shterenberg, Zhong Chen, and Keren Li. *Distributed Collaborative Learning with Representative Knowledge Sharing*. **Mathematics**, 2025, vol. 13, no. 6, 1004.
- [A.1] E. Chesnokova, D. J. Barrett, B. Geng, **Mengchen Fan**, T. Kulkarni, and S. Rothenberg. *An AI Physical Exam Can Predict Lung Function*. **American Thoracic Society (ATS) International Conference**, 2025, Poster Discussion Abstract, A5307.
- [C.6] **Mengchen Fan**, Baocheng Geng, Keren Li, Xueqian Wang, and Pramod K. Varshney. *Interpretable Data Fusion for Distributed Learning: A Representative Approach via Gradient Matching*. **International Conference on Information Fusion (FUSION)**, 2024, pp. 1–8.
- [J.3] Yusen Zhai, **Mengchen Fan**, Baocheng Geng, Xue Du, Scott Snyder, and Larrell Wilkinson. *Impact of Phased COVID-19 Vaccine Rollout on Anxiety and Depression among the U.S. Adult Population, January 2019–February 2023: A Population-Based Interrupted Time Series Analysis*. **The Lancet Regional Health – Americas**, 2024.
- [P.1] Jinju Qian, Zhichao Mo, Zhangquan Rao, Weibing Yue, Ke Wang, **Mengchen Fan**, Xiangyang Peng, Xiaofeng Pang, Lin Yi, and Rui Wang. *Routing Inspection Path Planning Method and Device for Routing Inspection Robot*. **Chinese Patent CN106156888B**, 2020.

MANUSCRIPTS SUBMITTED AND UNDER REVIEW

[†]Equal contribution

- [U.1] **Mengchen Fan**[†], Qizhen Lan[†], Xi Xiao, Lin Zhao, Liang He, Tianyang Wang, Xiaoqian Jiang, Pulin Che, and Baocheng Geng. *MedSTP: Preserving Subspace and Transition Evidence for Medical VQA Token Pruning*. **IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)**, 2027, under review.
- [U.2] Baraa Bilbeisi[†], **Mengchen Fan**[†], Baocheng Geng, and Qing Tian. *CRAD: Class-Wise Reliability-Aware Distillation for Decentralized Heterogeneous Federated Learning*. **IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)**, 2027, under review.

- [U.3] Qizhen Lan, Yiling Ma, Ruiqi Chen, **Mengchen Fan**, Xi Xiao, Lin Zhao, Liang He, Baocheng Geng, Pulin Che, Tianyang Wang, and Xiaoqian Jiang. *When Evidence Vanishes: Distilling Visual-Evidence Dependency in Medical Vision-Language Models*. **IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)**, 2027, under review.
- [U.4] Qizhen Lan, Xi Xiao, Xiangchen Guan, **Mengchen Fan**, Moule Lin, Jung Im Choi, and Lijing Zhu. *Trust Is Not Enough: Influence Calibration for On-Policy Self-Distillation in Agentic RL*. **AAAI Conference on Artificial Intelligence**, 2027, under review.
- [U.5] **Mengchen Fan**[†], Xu Zhang[†], Baocheng Geng, and Pulin Che. *SIGMA: Signaling via Golgi Morphology Analysis for Pathway-State Classification*. **Communications Biology**, 2026, submitted.
- [U.6] Xu Zhang[†], **Mengchen Fan**[†], Kim Seokoon, Qing Zhao, Brant Wagener, Baocheng Geng, and Pulin Che. *Machine Learning Classifies Subcellular Golgi Morphology from Imaging Flow Cytometry during Inflammatory Activation: An Exploratory Study*. **npj Imaging**, 2026, submitted.
- [U.7] Xu Zhang, **Mengchen Fan**, Xinnan Li, and Pulin Che. *Drug Repurposing from Single-Cell IPF Transcriptomes Identifies the HDAC Inhibitor Belinostat as an Antifibrotic Candidate*. **Respiratory Research**, 2026, submitted.
- [U.8] **Mengchen Fan**, Roman Shterenberg, and Keren Li. *A Statistical Framework for Measuring Data Heterogeneity in Distributed Learning*. **The Annals of Applied Statistics**, 2025, under review.

TEACHING AND MENTORING EXPERIENCE

University of Alabama at Birmingham

Spring 2026	Natural Language Processing CS 662/762 , Teaching Assistant, Guest Instructor
Spring 2026	Deep Learning CS 665/765 , Teaching Assistant, Guest Instructor
Fall 2025	Computer Security CS 436/636/736 , Teaching Assistant
Summer 2025	Mobile Application Development CS 422/522 , Teaching Assistant
Spring 2025	Deep Learning CS 665/765 , Teaching Assistant, Guest Instructor
Fall 2024	Foundations of Data Science CS 685 , Teaching Assistant
Summer 2024	Mobile Application Development CS 422/522 , Teaching Assistant
Spring 2024	Deep Learning CS 665/765 , Teaching Assistant, Guest Instructor
Fall 2023	Machine Learning CS 667/767 , Teaching Assistant
Summer 2023	Machine Learning CS 667/767 , Teaching Assistant, Guest Instructor
2025 – 2026	Student Mentoring: Baraa Bilbeisi , Federated learning, and knowledge distillation
2025 – 2026	Student Mentoring: Yifan Ying , Federated learning, and token merging

GRANT AND PROPOSAL CONTRIBUTIONS

- **UAB REINVENT Seed Funding Program** (\$50,000), 2025. *Machine Learning-Based Early Detection of Acute Respiratory Distress Syndrome* (funded). Proposal Contributor; developed the machine-learning methodology and contributed to proposal writing. PI: Pulin Che.

ACADEMIC SERVICE

- **Conference Reviewer:** AAAI (2026, 2027); MICCAI (2026); IJCNN (2025, 2027); ICME (2025); BioKDD (2025, External Reviewer).
- **Journal Reviewer:** IEEE/ACM Transactions on Computational Biology and Bioinformatics, 2024–2025; BMJ Open Ophthalmology, 2025; Journal of Clinical Pharmacy and Therapeutics, 2025; Frontiers in Genetics, 2025; Frontiers in Nutrition, 2025; Discover Oncology, 2025; Journal of Visualized Experiments (JoVE), 2026.

INVITED TALK

- **Distributed Model Pruning and Data Fusion** June 10, 2024
2024 WNAR/IMS/Graybill Annual Meeting, Colorado State University Fort Collins, CO, United States

PROFESSIONAL EXPERIENCE

- **Citi**

Dec 2020 – Aug 2022

Java Full Stack Developer

Shanghai, China

- Developed and maintained a full-stack SMA platform for Basel IV operational-risk calculations using Java, Spring, Angular, and Oracle; mentored new team members.

TECHNICAL SKILLS

Programming and Tools: Python, C++, Java, SQL, JavaScript; PyTorch, Spring, Angular, and Oracle.